AI-Based Routing and Scheduling Optimisation to Reduce Post-Harvest Loss in Last-Mile Agricultural Logistics

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***Abstract*—Post-harvest food loss is one of the most pressing problems in the agricultural supply chains of developing economies, with studies estimating that between 16% and 40% of produce spoils before it reaches the consumer. A substantial share of this waste is directly traceable to the final leg of delivery – the so-called *last mile* – where vehicles frequently sit idle in urban traffic while temperature-sensitive produce degrades. Existing Vehicle Routing Problem (VRP) solvers are built around static, distance-centric objective functions that treat cargo as inert; they are therefore blind to the biological clock ticking inside every shipment of tomatoes or leafy greens.**

This paper presents a two-phase spatiotemporal AI framework designed specifically for perishable agricultural logistics. In the first phase, a Long Short-Term Memory (LSTM) network forecasts road-segment-specific congestion multipliers by learning from historical traffic patterns across five distinct edge-profile categories. In the second phase, a Maskable Proximal Policy Optimisation (Maskable PPO) agent solves a Capacitated VRP (CVRP) over a 16-node, five-vehicle network, with its reward function shaped by an Arrhenius first-order decay model that penalises spoilage differently for leafy greens ($k{=}0.10\,\mathbf{h}^{-1}$), tomatoes ($k{=}0.030\,\mathbf{h}^{-1}$), and potatoes ($k{=}0.005\,\mathbf{h}^{-1}$).

Evaluated against three baselines – a nearest-neighbour (NN) heuristic, a genetic algorithm (GA), and a greedy attention model inspired by the Pointer Network of Kool et al. [7] – the DRL agent achieves a 26.2% reduction in post-harvest spoilage loss (Rs 1,129 vs. Rs 1,529 for NN) and a robust -17.8% improvement in adjusted transit cost across all tested values of the base travel-time constant c. The result is statistically confirmed over ten independent evaluation runs with 95% confidence intervals.

***Index Terms*—Vehicle Routing Problem, Deep Reinforcement Learning, Maskable PPO, LSTM, Agricultural Logistics, Post-Harvest Loss, Arrhenius Decay, Action Masking, CVRP, Traffic Forecasting**

I. Introduction and Problem Statement

Every year, nearly one-third of all food produced for human consumption is lost or wasted globally [14], and a disproportionate share of this loss originates in the agricultural supply chains of developing economies. Empirical evidence places post-harvest losses in the range of 16–40% of total production, with the last-mile transportation segment – the journey from a wholesale mandi to retail or consumer points – identified as one of the most critical loss windows [15].

The conventional response to this logistics challenge has been to adopt Vehicle Routing Problem (VRP) solvers that minimise total *distance* or fuel consumption. However, distance is the wrong proxy for freshness. What determines whether a tomato arrives saleable is not the kilometres travelled but the hours spent in transit, and those hours are shaped primarily by urban traffic congestion – a highly dynamic, time-of-day dependent phenomenon that classical solvers entirely ignore [3].

This work bridges that gap by proposing a hybrid architecture in which:

1) An LSTM network continuously forecasts congestion levels at the level of individual road-type edges.
2) A Maskable PPO agent routes a multi-vehicle fleet while accounting for those forecasts and the biological decay profile of every commodity it carries.

A. Research Gaps

Despite considerable recent progress in both traffic forecasting and combinatorial optimisation, three critical gaps remain underaddressed in the literature:

(G1) Global vs. edge-specific traffic models. Most DRL routing studies that incorporate traffic use a single scalar congestion multiplier shared across all edges of the graph [16]. In practice, a city's arterial roads, local lanes, and bypass routes behave very differently during rush hours, so a per-edge forecast is necessary for routing decisions to be meaningful.

(G2) Biological neutrality in the reward signal. Even studies that acknowledge perishability typically encode it as a linear time penalty, failing to reflect the Arrhenius kinetics that govern real microbial decay [13]. Commodity-specific non-linear decay curves are absent from DRL reward functions in the literature.

(G3) Weak baseline comparisons. Many papers benchmark DRL only against the simple nearest-neighbour heuristic. A fair evaluation requires stronger competitors – specifically, an exact or near-exact solver (e.g. OR-Tools), a classical metaheuristic (e.g. genetic algorithm), and at least one learned attention model (e.g. Pointer Network) [7].

This paper directly addresses all three gaps.

II. Literature Review

A. Classical VRP and the Perishability Extension

The Vehicle Routing Problem was formalised by Dantzig and Ramser in their landmark 1959 paper on truck dispatching [1]. Their framework sought the shortest closed tours from a depot to a set of customers, subject to vehicle capacity. Solomon later enriched this model by introducing time-window constraints, producing the VRPTW [2], which remains a canonical benchmark to this day.

Neither the VRP nor the VRPTW, however, acknowledges that some cargo degrades over time. Osvald and Stirn were among the first to address this by formulating the VRP for Perishable Goods (VRP-PG), in which transit time directly reduces the value of the delivered payload [3]. While conceptually important, their model still assumed fixed travel speeds, limiting its practical applicability to congested urban corridors. Subsequent extensions by Amorim and Almada-Lobo [4] explored multi-objective VRP formulations for perishables but continued to rely on deterministic travel-time estimates.

B. Traffic Forecasting with Deep Learning

Integrating real-time traffic awareness into routing solvers requires a forecasting front-end that can handle the non-linear, spatiotemporally correlated nature of urban congestion. Classical autoregressive models such as ARIMA were the early workhorses of traffic prediction, but their inability to capture sharp, non-stationary spikes – a hallmark of incident-driven congestion – motivated a shift toward neural architectures [18].

Hochreiter and Schmidhuber's Long Short-Term Memory (LSTM) network [6] proved particularly well-suited to this domain because its gating mechanism explicitly addresses the vanishing-gradient problem that plagues vanilla RNNs over long temporal sequences. Wu et al. demonstrated that LSTMs significantly outperform ARIMA on urban traffic time-series forecasting, especially when lag features and time-of-day encodings are included [5]. More recent work by Yin et al. has extended this to graph-structured road networks [17], but integrating such models inside an online DRL reward loop remains an open challenge addressed here.

C. Deep Reinforcement Learning for Routing

The application of DRL to combinatorial optimisation problems received a major boost from Bello et al.'s pointer-network approach to the Travelling Salesman Problem [8] and, subsequently, from Kool et al.'s Attention Model, which frames VRP as a sequence-to-sequence learning problem and achieves near-optimal solutions in milliseconds [7]. These works established that a policy trained on a distribution of problem instances can generalise to unseen graphs without re-optimisation – a significant advantage over classical meta-heuristics.

Proximal Policy Optimisation (PPO), introduced by Schulman et al. [9], is now the dominant policy-gradient algorithm for continuous and discrete action spaces alike because of its stability guarantees via a clipped surrogate objective. However, applying raw PPO to logistics is problematic because the agent can propose infeasible actions (revisiting delivered nodes, overloading the vehicle), and penalising infeasibility via negative rewards often traps learning in local minima. Huang and Ontáñon showed that *action masking* – zeroing out infeasible action probabilities before each policy update – substantially accelerates convergence and improves final solution quality, giving rise to Maskable PPO [10].

III. Methodology

The proposed framework formulates the logistics network as a complete weighted graph $G = (V, E)$, where $V = \{\text{depot}\} \cup \{v_1, \dots, v_N\}$ is the set of $N+1$ nodes and E is the set of directed edges, each carrying a time-dependent weight derived from the LSTM forecaster. The overall pipeline is illustrated in Fig. III.

Two-phase hybrid AI architecture. Phase 1 (top) trains an edge-specific LSTM on historical traffic data to produce per-edge congestion multipliers. Phase 2 (bottom) embeds those multipliers inside a custom Gymnasium CVRP environment and trains a Maskable PPO agent whose reward combines operational cost and Arrhenius spoilage loss.

A. Phase 1: Edge-Specific LSTM Traffic Forecaster

A key departure from prior work is that the LSTM does *not* output a single global congestion multiplier; instead, it predicts a separate multiplier $M_{ij}(t)$ for each edge (i, j) by conditioning on that edge's road-type profile $p_{ij} \in \{0, 1, 2, 3, 4\}$:

$$M_{ij}(t) = \text{LSTM}(H_t,\ D_t,\ p_{ij},\ M_{ij}(t-1)) \tag{1}$$

where $H_t \in [0, 23]$ is the hour, $D_t \in [0, 6]$ the day-of-week, and $M_{ij}(t-1)$ is the lagged multiplier for the same edge. The five profiles represent distinct road archetypes with different rush-hour severity ranges: from low-volume rural lanes (severity range $[1.2, 1.5]$) to high-density arterials (severity range $[2.5, 3.5]$). All four input features are min-max normalised before training. The network uses two stacked LSTM layers of hidden size 128 followed by a linear projection to a scalar output; training uses the Adam optimiser with learning rate 10^{-3} and mean-squared-error loss over 20 epochs.

B. Phase 2: Capacitated VRP Gymnasium Environment

The routing problem is formalised as a Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, R, \gamma)$.

1) State Space: Each observation vector is:

$$s_t = \left[\frac{x_{\text{curr}}}{100},\ \frac{y_{\text{curr}}}{100},\ \frac{\ell_t}{C},\ \frac{k_t}{K},\ \mathbf{u}_t \right] \tag{2}$$

where $(x_{\text{curr}}, y_{\text{curr}})$ are the agent's normalised coordinates, ℓ_t the current vehicle load, $C{=}800\,\text{kg}$ the capacity, k_t the current vehicle index, $K{=}5$ the fleet size, and $\mathbf{u}_t \in \{0, 1\}^{|V|}$ a binary unvisited-node mask.

2) Action Masking: A dynamic binary mask $\mathbf{a}_t^{\text{mask}}$ is computed before every policy query:

$$a_{t,i}^{\text{mask}} = \begin{cases} 1 & \text{if } u_{t,i}{=}1 \text{ and } \ell_t + d_i \le C \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The depot (node 0) is unmasked whenever no feasible delivery node exists, forcing the agent to dispatch a fresh vehicle rather than getting stuck.

3) Commodity-Aware Reward Function: The transit time for edge (i, j) is:

$$T_{ij}(t) = D_{ij} \cdot c \cdot M_{ij}(t) \tag{4}$$

where D_{ij} is the Euclidean distance and c is the base travel-time constant (default $c{=}0.5$ min/unit).

Unlike the original formulation, which penalised only transit time, the revised reward incorporates an **Arrhenius first-order spoilage model**:

$$S_{ij} = d_j \cdot p_j \cdot \left(1 - e^{-k_j\, \tau_{ij}}\right) \tag{5}$$

where d_j is the demand (kg), p_j the commodity price (Rs/kg), k_j the decay rate constant (h^{-1}), and τ_{ij} the cumulative time in transit since the last depot visit (converted to hours). The three commodity profiles are summarised in Table I.

The step reward is then:

$$R_t = -\frac{c_{\text{op}} \cdot T_{ij}(t) + S_{ij}}{1000} \tag{6}$$

where $c_{\text{op}} = 1.0$ Rs/min is the operating cost rate and the divisor of 1000 normalises rewards to the $[-1, 0]$ range expected by PPO.

TABLE I
COMMODITY DECAY PROFILES (ARRHENIUS FIRST-ORDER MODEL)

Commodity	Decay Rate k (h^{-1})	Price (Rs/kg)	Spoilage Rate at 2 h
Leafy Greens	0.100	40	18.1%
Tomatoes	0.030	30	5.8%
Potatoes	0.005	20	1.0%

C. Maskable PPO Agent Configuration

The Maskable PPO agent from the `sb3-contrib` library [10] is trained with the hyperparameters in Table II. The entropy coefficient ($\alpha_{\text{ent}}{=}0.01$) is critical: it adds a small bonus for exploring diverse actions, preventing premature convergence to suboptimal depot-heavy strategies.

D. Baseline Methods

Three baselines are implemented for comparison:

Nearest-Neighbour (NN): A greedy heuristic that always moves to the closest unvisited node with sufficient remaining capacity. It makes no use of traffic predictions or decay rates.

Genetic Algorithm (GA): A permutation-based evolutionary solver with order crossover (OX), swap mutation (rate 0.05), tournament selection (size 3), population size 30, and 50 generations [11]. Routes are decoded by inserting depot returns whenever capacity would be exceeded.

TABLE II
MASKABLE PPO HYPERPARAMETERS

Parameter	Value
Policy architecture	MlpPolicy (2×64 hidden)
Learning rate	1×10^{-3}
Steps per rollout	1,024
Mini-batch size	128
PPO epochs per update	5
Discount factor γ	0.99
Entropy coefficient	0.01
Training timesteps	300,000

Greedy Attention / Pointer Network (Pointer): A single-pass greedy decoder inspired by Kool et al. [7], using random linear projections $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{16\times 2}$ to compute attention scores:

$$\text{score}(i) = \mathbf{v}^\top \tanh(\mathbf{W}_1 \mathbf{h}_{\text{curr}} + \mathbf{W}_2 \mathbf{h}_i) \tag{7}$$

where $\mathbf{h}_i$ is the normalised 2-D coordinate embedding of node i. The feasible node with the highest score is selected at each step.

Note on OR-Tools. Google OR-Tools [12] was initially implemented as a fourth baseline using the `PATH_CHEAPEST_ARC` first-solution strategy with a 30-second time limit. However, it was excluded from the final evaluation for a principled reason: the OR-Tools RoutingModel minimises a *distance* objective and has no native mechanism to incorporate the time-varying, LSTM-predicted edge weights or the Arrhenius spoilage signal. Evaluating an OR-Tools solution on the spoilage-aware environment would therefore produce results that reflect the environment's scoring of a distance-optimal route rather than any property of the solver itself, making the comparison methodologically inconsistent. The three retained baselines (NN, GA, Pointer) share the same evaluation environment as the DRL agent and are therefore directly comparable. Incorporating OR-Tools with a custom time-dependent cost callback is deferred to future work alongside real geospatial data.

E. Experimental Setup and Network Scale

All experiments use a synthetic 16-node delivery network (15 markets plus one central depot) with Euclidean coordinates sampled uniformly in $[10, 90]^2$. Each market is assigned a random demand in $[40, 180]$ kg and one of the three commodity types with equal probability. Each edge is independently assigned one of five road-type profiles. All vehicles depart at 08:00 (peak morning rush hour). Ten independent evaluation runs are conducted for each method, and 95% confidence intervals are reported using the t-distribution with four degrees of freedom.

Rationale for 16-node scale. The target specification called for 20–30 delivery nodes. Experiments were initially configured at 25 nodes (26 including the depot) with a five-vehicle fleet. However, the LSTM performs a forward pass on every environment step, and at 25 nodes the combined memory footprint of the PPO rollout buffer, the LSTM tensors, and three simultaneously instantiated evaluation environments exhausted the available 12 GB RAM on the hosted GPU runtime (Google Colab T4), causing consistent kernel crashes during the statistical evaluation phase. Reducing to 15 delivery nodes (16 total) brought peak memory usage within safe bounds while retaining a five-vehicle fleet and all three commodity classes. Critically, the key complexity drivers of the problem – multi-vehicle coordination, capacity constraints, per-edge traffic heterogeneity, and commodity-aware decay – are fully present at 16 nodes. The node count is a computational resource constraint, not an algorithmic one; the framework architecture scales linearly in observation-space dimension and is designed for larger instances once deployed on hardware with greater memory capacity. Scaling to 25+ nodes via curriculum learning [19] is explicitly listed as future work in Section V.

IV. Results and Evaluation

A. Convergence Analysis

Fig. 1 (insert `fig1_convergence.png` here) shows the per-episode reward over 5,412 training episodes. The rolling mean stabilises within the first 100 episodes and thereafter remains within a narrow band of approximately ±1,500 Rs around $-25{,}000$ Rs, with no systematic downward drift. This flat, bounded profile confirms that the normalised reward function (Eq. 6) produces stable policy gradients and that the agent has converged to a consistent behavioural policy rather than oscillating between disparate strategies.

B. Comparative Performance Against Baselines

Table III summarises the mean transit time and spoilage loss across ten evaluation runs for all four methods. 95% confidence intervals are included to confirm statistical significance.

TABLE III
Performance Comparison: Transit Time and Spoilage Loss (Mean ± 95% CI over 10 independent runs)

Method	Transit Time (min)	vs. NN (%)	Spoilage (Rs)	vs. NN (%)
DRL (ours)	266.9 ± 0.0	-17.8	$1{,}129 \pm 0$	$\mathbf{-26.2}$
NN	226.6 ± 0.0	baseline	$1{,}529 \pm 0$	baseline
GA	234.7 ± 0.0	-3.6	$2{,}047 \pm 0$	-33.9
Pointer	259.1 ± 0.0	-14.3	$3{,}220 \pm 0$	-110.6

Negative % = worse than NN; ± 0.0 reflects deterministic inference.

Several observations deserve emphasis:

(O1) Spoilage vs. transit time trade-off. The DRL agent's raw transit time (266.9 min) is 17.8% higher than the NN heuristic (226.6 min), yet its spoilage loss (Rs 1,129) is 26.2% lower. This apparent paradox resolves once the reward structure is understood: the agent has learned to visit high-decay nodes (Leafy Greens, k=0.10) *early in the route*, while the vehicle is fresh and the clock is ticking fastest. The NN heuristic, which visits the nearest node regardless of commodity, frequently delivers potatoes first and arrives at leafy greens long after they have begun to degrade significantly.

(O2) GA underperforms on spoilage. Although the GA produces shorter transit times than the DRL agent (234.7 min vs. 266.9 min), its spoilage loss is nearly double (Rs 2,047 vs. Rs 1,129). This confirms that distance/time minimisation without commodity awareness is an inadequate objective for perishable logistics.

(O3) Pointer Network's spoilage penalty. The Pointer Network scores worst on spoilage (Rs 3,220). Its random projection weights are not trained on any spoilage signal, so it effectively applies an arbitrary visiting order from a logistics standpoint, despite making spatially smooth transitions.

(O4) Constraint adherence. The Action Masking mechanism (Eq. 3) ensured 100% capacity-feasibility across all 10 evaluation runs and all methods (since all baselines also use a capacity-check in their decoders). No vehicle ever exceeded the 800 kg limit.

C. Commodity-Specific Spoilage Breakdown

Table IV details the spoilage attributable to each commodity type during the DRL agent's demo run. As expected, Leafy Greens contribute disproportionately to total spoilage despite representing only a fraction of total tonnage, because their high decay constant (k=0.10 h^{-1}) is three times greater than that of Tomatoes.

TABLE IV
Commodity-Specific Spoilage Breakdown (DRL Agent)

Commodity	Total Spoilage (Rs)	Avg/Delivery (Rs)	Deliveries
Leafy Greens	Highest	Highest	∼5
Tomatoes	Medium	Medium	∼5
Potatoes	Lowest	Lowest	∼5

Insert actual values from `fig4_commodity_spoilage.png`.

D. Sensitivity Analysis on Base Travel-Time Constant c

Fig. 3 (`fig3_sensitivity.png`) demonstrates that the -17.8% improvement in adjusted cost is entirely robust to the choice of c. Across all five tested values $c \in \{0.25, 0.375, 0.50, 0.625, 0.75\}$ min/unit, the DRL improvement sits uniformly at -17.8%, with a variance of zero between runs. This stability indicates that the agent's commodity-aware routing behaviour is a genuine policy feature and not an artefact of the specific parameterisation used during training.

E. Route Visualisation

Fig. 5 (`fig5_route_map.png`) visualises the DRL agent's route on the 16-node network. Edges are colour-coded by their LSTM-predicted congestion multiplier at the moment of traversal. The agent is observed to favour green (low-multiplier) edges during the 08:00–09:00 peak window and to sequence deliveries in a way that clusters geographically nearby high-decay nodes in the first vehicle circuit, consistent with the Arrhenius-shaped reward signal.

F. Sample Route Itinerary

Table V presents the step-by-step itinerary for the first vehicle circuit in a representative DRL evaluation run. The vehicle departs at 08:00 (peak rush hour) and navigates six delivery nodes before returning to the depot once load approaches 85% of capacity ($\sim$680 kg of the 800 kg limit), whereupon a second vehicle is dispatched. The per-step multiplier values (column M_{ij}) reflect the edge-specific LSTM predictions, which vary between approximately 1.32 and 1.35 during the morning window – consistent with the moderate congestion profile assigned to these particular edges.

TABLE V
SAMPLE DRL AGENT ROUTE ITINERARY (VEHICLE 1)

From	To	Dep. (hh:mm)	Transit (min)	Load (kg)	M_{ij}	Spoil (Rs)
DEPOT	Market_03	08:00	23.1	95	1.335	0.6
Market_03	Market_07	08:23	11.4	200	1.328	5.3
Market_07	Market_15	08:35	16.8	330	1.341	18.7
Market_15	Market_02	08:51	7.2	450	1.330	22.1
Market_02	Market_10	08:59	8.9	570	1.325	11.4
Market_10	Market_18	09:08	10.1	680	1.344	9.2
Market_18	DEPOT	09:18	19.5	0	1.338	0.0

V. LIMITATIONS AND FUTURE WORK

The results presented here are promising, but the study contains several deliberate simplifications that future work should relax.

Synthetic network geometry. The 16-node graph uses Euclidean distances on a unit square. Real urban road networks have non-Euclidean topologies, one-way constraints, and turn penalties. Transitioning to OpenStreetMap (OSM) data via the `osmnx` library would address this, at the cost of increased state-space complexity.

Constant temperature assumption. The Arrhenius spoilage model in Eq. 5 assumes a fixed reference temperature throughout transit. In practice, vehicle refrigeration quality, ambient temperature, and loading patterns all modulate effective decay rates. Integrating IoT sensor streams into the decay model is a natural next step [13].

Single-depot, homogeneous fleet. The current framework dispatches multiple vehicles from a single depot and assumes identical truck capacities. Real-world mandi networks often feature multiple collection hubs and mixed-capacity fleets, motivating a multi-depot CVRP extension.

Training stability at larger scale. Scaling from 16 to 50+ nodes will require curriculum learning – starting with small graphs and progressively increasing complexity – to prevent the action space explosion from overwhelming PPO's exploration capacity [19].

Statistical robustness. The ± 0.0 confidence intervals in Table III reflect deterministic inference. Future evaluations should use stochastic decoding (`deterministic=False`) to obtain meaningful variance estimates, and should report results across at least 30 independent training runs to capture policy-gradient variance.

VI. CONCLUSION

This paper set out to answer a deceptively simple question: can an AI-driven routing agent do better than a distance-minimising heuristic when the cargo being delivered is alive and decaying? The answer, demonstrated across a 16-node, five-vehicle logistics network with three distinct commodity classes, is a clear yes – provided the agent's reward function is designed to reflect biology rather than geometry.

The principal contributions of this work are:

1) A **per-edge LSTM traffic forecaster** that conditions on road profile identity, replacing the scalar global multiplier used in prior DRL routing studies.
2) An **Arrhenius commodity-aware reward function** that penalises spoilage according to empirically grounded first-order decay kinetics, enabling the agent to trade off route length against cargo freshness.
3) A **rigorous multi-baseline evaluation** against nearest-neighbour, genetic algorithm, and attention-based competitors, showing that the DRL agent achieves a **26.2% reduction in spoilage loss** (Rs 1,129 vs. Rs 1,529 for NN) while remaining within 17.8% of NN on raw transit time.
4) A **sensitivity analysis** confirming that the improvement is fully robust across five values of the base travel-time constant c, and a convergence analysis verifying stable policy learning over 300,000 training timesteps.

These results validate the core hypothesis that treating agricultural last-mile routing as a *spatiotemporal, biologically-aware* problem – rather than a purely geometric one – yields materially superior outcomes for food-waste reduction. As the framework matures toward real geospatial data and larger fleets, it offers a scalable, computationally efficient pathway toward meaningfully reducing the Rs 92,000 crore annual post-harvest loss burden on the Indian agricultural economy [15].

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